

Sahakari Finance Application

Functionality Report and Comparison Checklist

Purpose

This document summarizes the functionality currently implemented in the Sahakari cooperative finance application, how key workflows operate, what automation and reporting exist, and which production-grade features still need comparison against a real cooperative organization.

Field	Value
Report date	2026-06-06
Application type	Local-network cooperative finance / sahakari operations system
Frontend	React + Vite desktop operations UI
Backend	FastAPI, SQLAlchemy, PostgreSQL-ready models, JWT/RBAC dependencies
Current status	Functional foundation with expanded operations UI; not yet complete for live regulated production

1. Executive Summary

Current position

The app is no longer only a small registration form. It now includes member onboarding, scalable member search, member 360 profile, shares, savings, cash counter, loans, repayments, accounting, rules, automation, reports, approval queues, Excel-ready export, security/compliance screens, and dashboard metrics.

- Suitable as a working prototype/foundation for a cooperative finance office.
- Backend includes members, nominees, shares, savings, loans, installments, payments, accounts, journals, ledgers, closings, scheduler jobs, calculations and audit logs.
- Frontend exposes operational workspaces instead of only a registration form.
- Main remaining gaps: full maker-checker approval, locked-period enforcement, richer product masters, true XLSX import/export, backup/restore UI, cashier session depth and formal financial statements.

2. Implemented Functionality Matrix

Module	What it does now	Status
Authentication	Staff login using email/password and JWT token storage.	Implemented
Dashboard	Members, savings, loan exposure, risk, cash movement, pending KYC, high-risk cases.	Implemented
Member onboarding	Detailed person form: KYC, address, documents, family, guardian, nominee, occupation, income, risk flags, media references.	Implemented
Member search	Server-side search by member number, name, or phone with skip/limit paging and total count.	Implemented
Member 360 profile	Member data, nominees, shares, share activity, savings accounts, savings ledger, loans, schedules, payments, calculations.	Implemented
Share capital	Share purchase posting and share register display.	Implemented
Savings accounts	Open savings account, list active balances, post deposit and withdrawal.	Implemented
Cash counter	Deposit and withdrawal screens connected to savings transactions and ledger posting.	Implemented
Loans	Loan disbursement, flat installment schedule generation, loan portfolio, schedule view, installment payment.	Implemented
Accounting	Chart of accounts and trial balance endpoints/UI.	Implemented
Rules engine	Interest, penalty, accrual rule CRUD/read views; calculation history.	Implemented foundation
Automation	Manual automation run, daily interest trigger, overdue marking trigger, scheduler job list, calculation results.	Implemented foundation

Module	What it does now	Status
Reports	Portfolio summary, daily operations, approval queue, high-risk queue, overdue queue, trial balance.	Implemented foundation
Excel/export	CSV member portfolio export, openable in Excel.	Implemented basic
Security/compliance	RBAC dependencies, audit logging in services, security workspace, risk flags.	Implemented foundation
Closings	Day/month/year closing models and closing register display.	Foundation only

3. How Key Workflows Work

Workflow	Operational behavior
Member onboarding	Staff enters personal, KYC, document, address, family, nominee, occupation, income and risk information. Backend stores the member, JSON profile sections, nominee records and audit entry.
Finding one member among 10k+	Staff searches by member number, name or phone. Backend applies indexed search and returns paged results. Staff clicks Open to load the member 360 profile instead of scrolling thousands of rows.
Member 360 review	The profile endpoint gathers the selected member's shares, savings accounts, savings transactions, loans, installments, payments, nominees and calculation results into one operational view.
Savings deposit	Backend locks the account, increases balance, creates savings transaction, posts double-entry ledger lines, and writes audit log.
Savings withdrawal	Backend checks active account and sufficient balance, decreases balance, posts transaction, ledger and audit.
Loan disbursement	Backend creates active loan, generates flat monthly installments, posts loan disbursement ledger entry and audit log.
Loan repayment	Staff opens loan schedule and pays an installment. Backend marks installment paid, reduces outstanding principal, posts principal and interest ledger entries, and audits the payment.
Automation run	Admin/manager runs automation for a date. Backend calculates savings interest and overdue penalties through the rules engine, updates live records, stores calculation results and returns posting summary.
Reports/export	Reports screen loads portfolio totals, queues and trial balance. CSV export downloads member portfolio data for Excel review.

4. Current Screens and API Coverage

Frontend Workspaces

Screen	Main capability
Dashboard	Daily snapshot, quick actions, recent members, health signals.
Members	Onboarding, search/paging, member 360 profile.
Shares	Share purchase and share register.
Savings	Open account and view account balances.
Cash Counter	Post deposit/withdrawal and monitor counter balances.
Loans	Disburse loan, view portfolio, open schedule, post repayment.
Reports	Portfolio, approval queue, high-risk cases, overdue queue, trial balance, CSV export.
Automation	Daily/monthly/fiscal automation buttons, scheduler jobs, closing register, calculations.
Rules	Create interest rule and inspect interest/penalty/accrual rules.
Accounting	Chart of accounts and trial balance.
Security	Compliance controls, RBAC/audit/export/backup guidance.

Backend API Groups

- /auth: login
- /dashboard: operational metrics
- /branches: branch list/create
- /members: list, count, profile, create, update, nominees
- /shares: list and purchase
- /savings: accounts, deposit, withdrawal
- /loans: list, create, installments, payments
- /accounting: chart of accounts, trial balance
- /reports: portfolio summary, daily operations, approval queue, CSV export
- /automation: manual run and async triggers
- /rules: interest, penalty, accrual, scheduler jobs, calculation results

5. Calculations, Ledgers and Automation

Area	Current behavior	Notes
Savings interest	Rules engine checks active interest rule by savings product/account type; falls back to account interest rate.	Foundation implemented; product-level configuration should be expanded.
Penalty	Rules engine calculates overdue installment penalty from configured rule or fallback rate.	Supports fixed/percentage/compound concepts in rule schema.

Area	Current behavior	Notes
Loan schedule	Flat monthly principal plus monthly interest schedule generated at disbursement.	EMI/reducing/diminishing variants are not yet fully implemented.
Double-entry ledger	Savings, shares, loan disbursement and repayment services post journal/ledger entries.	Needs stricter balancing audits and approval before live use.
Calculation records	Automation stores calculation_results for traceability.	Good foundation for explainable historical calculations.
Closings	Models exist for day, month and year closings; UI displays closing register.	Actual close/lock workflow needs more implementation.

6. Excel and Reporting

Excel recommendation

Excel should be used for exports, imports, migration, audit sharing and offline review. It should not be used as the primary database for live cooperative finance records.

- Available now: CSV member portfolio export at `/api/v1/reports/portfolio-export.csv`.
- CSV can be opened in Microsoft Excel and used for comparison or audit review.
- Recommended next step: true .xlsx exports with multiple sheets for members, savings, loans, shares, trial balance and daily cash.
- Recommended next step: bulk import with validation preview for opening balances and member migration.

7. Production Gap Checklist

Feature	Expected in real production	Current
Maker-checker approval	Formal approval screens for KYC, loan, withdrawal, adjustment, reversal, closing.	Missing
Locked-period enforcement	Reject edits/postings after day/month/year close unless reversal workflow is used.	Mostly missing
Product masters	Savings products, FD products, loan products, share products, fee products.	Missing
Advanced loan engine	EMI, reducing balance, diminishing, partial payment, reschedule, collateral, guarantors.	Partial
Cashier session	Opening cash, denomination count, handover, vault transfer, shortage/excess handling.	Partial
Backup/restore UI	Encrypted backups, restore validation, backup schedule and status dashboard.	Missing

Feature	Expected in real production	Current
True XLSX reports	Formatted multi-sheet Excel export and import validation.	Basic CSV only
Financial statements	Balance sheet, profit/loss, ledger statement, cash book, member statement, regulatory reports.	Partial
Notifications	SMS/email/reminder outbox for due loans, overdue, maturity, KYC expiry.	Missing
Session/rate limit	Refresh tokens, session tracking, login rate limiting.	Partial
Branch scope	Strict branch-scoped data access for all modules.	Partial
Audit viewer	Searchable audit log UI with filters and export.	Missing UI
Backup of documents/media	Real file upload/storage for photo, signature, thumbprint and KYC files.	Reference fields only

8. Real-Organization Comparison Template

Area to compare	Questions for the cooperative office	Mark
Membership	What exact KYC fields, documents, nominees, guardian rules and membership types are required?	
Savings	Which account products exist, interest methods, withdrawal limits, dormancy rules and fees?	
Loans	What loan products, approval stages, guarantors, collateral, interest methods and penalty rules are used?	
Shares	Share purchase limits, transfer/refund rules, certificate format and dividend process?	
Cash counter	How many counters, cashier opening/closing process, vault movement and denomination requirements?	
Accounting	Chart of accounts, voucher approval, fiscal year close, reserves and dividend postings?	
Reports	Daily, monthly, annual, regulatory, board and member statement reports required?	
Excel	Which data must be exported/imported, and what format does the office currently use?	
Security	Roles, permissions, branch access, audit review and password/session policies?	
Backup	Backup location, frequency, encryption, external drive/NAS and restore testing policy?	

9. Recommended Next Development Phases

Phase	Goal	Deliverables
Phase 1	Approval and control	Maker-checker queues, approval actions, locked-period checks, reversal workflow.
Phase 2	Product depth	Savings/FD/loan product masters, interest methods, fees, collateral and guarantors.
Phase 3	Cash operations	Cashier sessions, vault movement, denominations, handover and day close.
Phase 4	Reports and Excel	Member statement, passbook, cash book, balance sheet, P&L;, XLSX export/import.
Phase 5	Backup and production hardening	Encrypted backups, restore UI, audit viewer, session/rate-limit, branch scope tests.